

Review

Prediction of Gd(III) complex thermodynamic stability

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ABSTRACT

Gadolinium(III) complexes are widely employed as contrast agents in diagnostic magnetic resonance imaging, and the development of new contrast agents remains a highly active field. A key requisite for contrast agents is high thermodynamic stability of the complex to ensure that the Gd(III) ion is not released in the body. Here we utilized published stability constants spanning 20 orders of magnitude to develop empirical expressions based on structural descriptors to predict the Gd(III)-ligand formation constant as well as pGd at pH 7.4. We then tested the predictive power of these expressions and found excellent agreement with a mean deviation of 1.0 log *K* units. The magnitudes of the structural descriptors are useful for guiding ligand design for Gd(III), and the empirical stability constant expressions can be used to screen potential ligands. This methodology is readily extended to other aqueous metal ion systems.

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1. Introduction

Gadolinium(III) complexes with polyaminopolycarboxylate ligands have been intensively investigated in the last three decades

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due to the application of some of them as contrast agents in clinical magnetic resonance imaging (MRI) [1–6]. Commercially available contrast agents are small Gd(III) chelates with octadentate ligands that leave a coordination position available for a water ligand. The coordinated water molecule is involved in chemical exchange with the water of tissues and organs, which provides an efficient mechanism for the shortening of longitudinal and transverse relaxation times of water proton nuclei. This relaxation enhancement effect can be used to increase the image contrast by applying adequate pulse sequences [6].

A key aspect of the Gd(III) complexes used as contrast agents is their stability, which must be high enough to prevent their dissociation within the body and the release of the toxic metal ion. While Gd(III)-based contrast agents are being used routinely in clinical practice and biomedical research, some toxicity issues have been identified [7]. In 2006, a potentially fatal disease named nephrogenic systemic fibrosis was associated with the administration of Gd(III) contrast agents to patients with renal impairment [8,9]. More recently, increasing evidence for the accumulation and retention of Gd(III) in the brain and other tissues has been reported, although it is not clear whether the deposited metal ion is still present as the chelate or in another chemical form [10,11]. The biological ramifications of this retained Gd(III) are as yet unknown [12,13].

There has been a major effort to improve the properties of the commercially available contrast agents by introducing different modifications to the ligand scaffold and developing innovative ligand structures [14,15]. As a result, the coordination chemistry of Gd(III) and the other lanthanide ions has experienced an enormous development, thanks to the combined research effort of many groups [1,16,17].

There is a wide consensus in the scientific community that the integrity of Gd(III) complexes *in vivo* is related both to kinetic inertness and thermodynamic stability [18]. Contrast agents must present a high thermodynamic stability and, perhaps more importantly, they must be inert with respect to their dissociation [19]. While inertness is clearly an important factor when considering the initial biological exposure to a contrast agent, thermodynamic stability will play a major role when considering the long term (months to years) fate of these complexes in the body, particularly in the case of linear contrast agents. Thermodynamic stability data of many Gd(III) complexes was accumulated during the course of the last two decades, particularly for complexes structurally related to the commercial contrast agents $[\text{Gd}(\text{DTPA})(\text{H}_2\text{O})]^{2-}$ and $[\text{Gd}(\text{DOTA})(\text{H}_2\text{O})]^-$. For these systems, a wide range of structural modifications were introduced to the ligand scaffold and the impact of such modifications in complex stability was analyzed. These thermodynamic data allowed for the establishment of some general principles for the design of stable Gd(III) complexes.

Here we sought to use the large body of thermodynamic data to develop a quantitative analysis of the thermodynamic stability constants on the basis of structural descriptors. We show that the thermodynamic stability of Gd(III) complexes can be predicted to a good accuracy using an empirical expression relying on structural descriptors. This correlation provides a powerful tool to aid ligand design and rationalize the different factors that determine the stability of Gd(III) complexes in solution. To validate the approach we applied this methodology to 12 new Gd(III) complexes. A quantitative structure–property relationship based on a machine-learning method was recently developed to predict the thermodynamic stability constants of Gd(III) complexes ($\log K_{\text{GdL}}$) [20]. This method relied on parametrized functions encoding the 2D structure of the ligand. Another attempt to correlate the structures of Gd(III) complexes and their stability constants used topological and quantum chemical descriptors to describe the

structures of the ligands of a small data set of about 20 compounds [21]. Our approach is based on structural descriptors that can be easily identified, such as the number and type of donor groups and the size of the macrocyclic platform, which represents a potentially more chemically intuitive way to design new chelators.

2. Stability constants of Gd(III) complexes

2.1. Complexes with acyclic ligands

The stability constants of a rather large number of Gd(III) complexes with acyclic ligands have been reported in the literature. Particularly exhaustive was the investigation of H_5DTPA and H_3DTPA -bis(amide) derivatives, which is clearly attributable to the use of these complexes as contrast agents in clinical practice. A first set of complexes that has been investigated is comprised by H_5DTPA (**A1** [22–24]) and derivatives containing different substituents at the central N atom (Fig. 1). The nature of the pendant arm attached to the central N atom was varied to introduce an α -substituted acetic acid arm (**A2–A4**) [25,26], alkyl groups (**A5, A6**) [27,28], or methylphosphonic acid (**A7, A8**) [29], acetamide (**A9**) [23,30] and methylpyridine functions (**A10**) [31]. Among these substituents, only the incorporation of a methylphosphonic acid group results in a complex with higher stability than $[\text{Gd}(\text{DTPA})(\text{H}_2\text{O})]^{2-}$ itself (Table 1).

Another group of H_5DTPA derivatives contains two substituents on the terminal N atoms that replace the acetic acid groups of the former (Fig. 1). This family is largely dominated by H_3DTPA -bis(amide) ligands (**A14–A23**) [23,32–36], but ligands containing α -substituted acetic acid (**A11**) [25], acetic acid ester (**A12**) [23], alkyl (**A13**) [37], and methylpyridine (**A24**) groups [32], were also investigated. In the case of bis(amide) derivatives the inspection of the stability constants (Table 1) shows that ligands with primary amide groups (**A14**) provide complexes with lower stability than those containing secondary amides (**A15–A20**), while ligands containing tertiary amide groups (**A21–A23**) show the highest stability. The stability constants determined for H_5DTPA (**A1**) and H_3DTPA -BMA (**A16**) using different ionic strengths do not differ significantly (Table 1).

A third group of H_5DTPA derivatives contain a substituent other than acetic acid on only one of the terminal N atoms (**A25–A29**, Fig. 1) [22,23,30,38], or three substituents other than acetic acid on each of the three N atoms of the H_5DTPA scaffold (**A30**) [25]. The complex of **A28** contains a propionate group replacing one of the terminal acetic acid arms of H_5DTPA , a substitution that decreases the stability constant by more than 2 log K units [38]. Finally, the stability constants of H_5DTPA derivatives containing a C-functionalized scaffold, such as MS-325 (Gd**A31**) and $\text{H}_5\text{EOB-DTPA}$ (**A32**) have been also reported [39,40]. The stability constants of these complexes are very similar to that of the parent H_5DTPA complex, which indicates that C-functionalization has a minor impact in complex stability.

The family of H_5DTPA derivatives **A1–A32** present stability constants that cover a range of approximately eight orders of magnitude, with the complex of the bis(amide) derivative **A14** showing the lowest stability constant ($\log K_{\text{GdL}} = 14.6$), and that of **A7**, which contains a methylphosphonic acid group, yielding the highest stability constant ($\log K_{\text{GdL}} = 22.79$) [29,33].

Besides H_5DTPA derivatives, thermodynamic stability constants of complexes with linear, structurally related ligands, have been also reported. Some of these ligands are shown in Fig. 2, while the corresponding stability constants are compiled in Table 1. This group of ligands contain one or more of the following structural modifications: 1) Propylene chains replacing one (**A34, A35**) or two (**A33, A38**) of the ethylene bridges of H_5DTPA derivatives

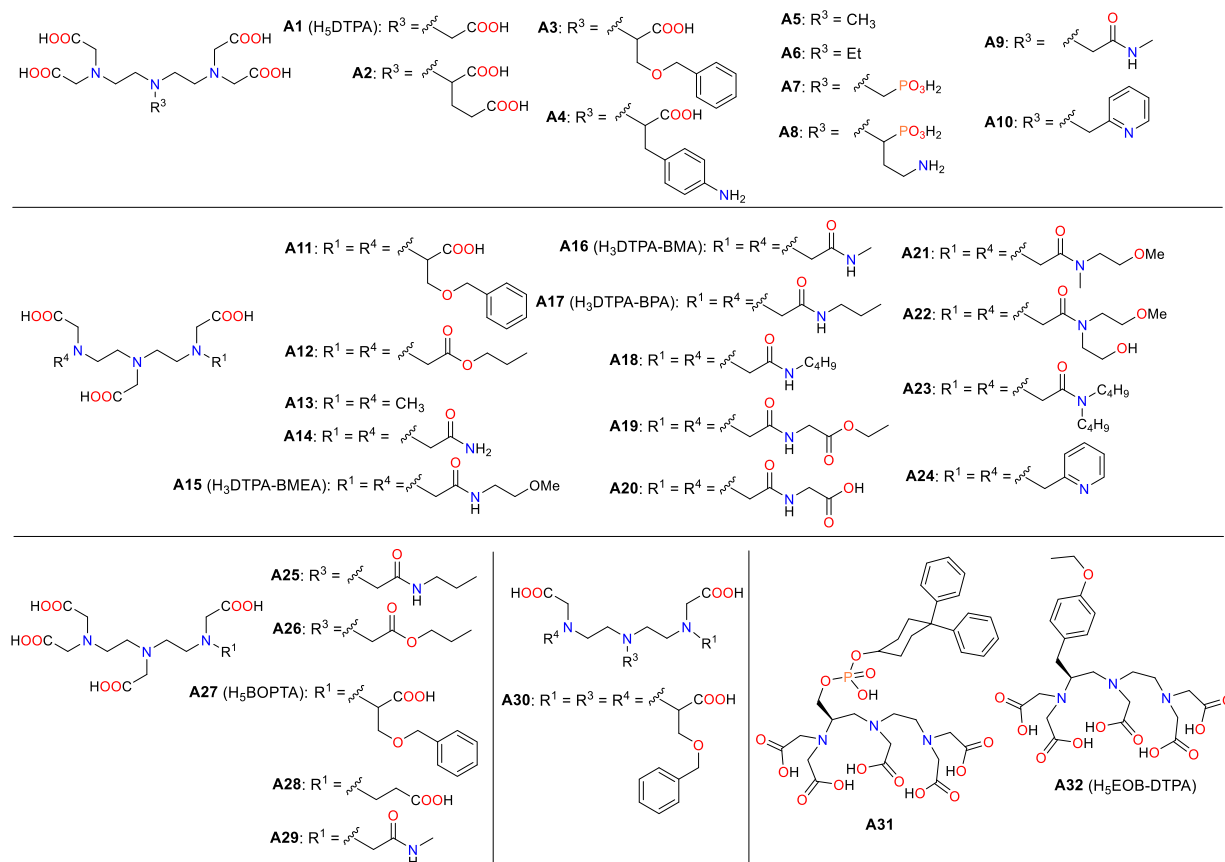


Fig. 1. Structures of different H₅DTPA derivatives.

[67,68,72]; 2) Ether oxygen atoms instead of the amine N atoms of the H₅DTPA backbone (**A36–A38**, **A40**, **A41**) [42–44,71]; 3) A pyridyl group replacing the central amine N atom (**A39**) [41]; 4) Phenyl spacers instead of the ethylene bridges of H₅DTPA (H₄BAPTA, **A41**) [44]; 5) Extended linear structures with a fourth donor atom in the poly(oxo)amine chain (**A40–A42**) [45,71,72]. These structural alterations result in a broad range of thermodynamic stability constants, the lowest being that of the complex of H₄BAPTA (**A41**, log K_{GDL} = 10.6) [44], and the highest that of the H₆TTHA complex (**A42**, log K_{GDL} = 28.4) [45]. This expands the range of stability constants of H₅DTPA derivatives to up to 18 orders of magnitude.

Another important type of non-macrocyclic ligand is H₄EDTA (**A43**, Fig. 3) [46], and a wide family of derivatives that has been extensively investigated. The central ethylenediamine unit has been functionalized with different groups, including acetamides (**A44–A46**) [47,48], methylpyridine (**A47**, **A54**) [49,55], phenol (**A49**, **A53**) [50,54] or a hydroxyethyl group (**A55**) [56]. Ligands with lower denticity in which some of the acetic acid groups are either missing (**A48**) or are replaced by alkyl substituents (**A51**, **A52**) can be also included in this group [46,52,53]. One of the systems (H₄OCTAPA, **A50**) contains bidentate 6-methylpicolinic acid groups [51], a binding motif that has been incorporated to a wide range of macrocyclic ligands (see below). Different structural modifications have been also incorporated to the H₄EDTA scaffold (Fig. 3), by substituting the ethylene linker by cyclohexyl (**A56** (H₄CDTA), **A57** (H₄CHXOCTAPA)) [46,57], propyl (**A58** (H₄TMDTA), **A59**) [58], phenyl (**A60**, H₄PhDTA) [59] or cyclobutyl (**A69**, **A70**, **A71**) spacers [65]. Ligand **A61** can be seen as a tripodal ligand, but also as an H₄EDTA derivative in which one of the N atoms is replaced by an ether oxygen atom and two acetic acid pendant

arms are missing [25]. The stability constants of H₄EDTA derivatives also cover a wide range of almost 13 orders of magnitude. The highest stability constant was reported for the complex with the octadentate ligand **A57** (H₄CHXOCTAPA, log K_{GdL} = 20.68) [57], while the lowest corresponds to the complex with the tetradentate ligand **A61** (log K_{GdL} = 7.88) [25].

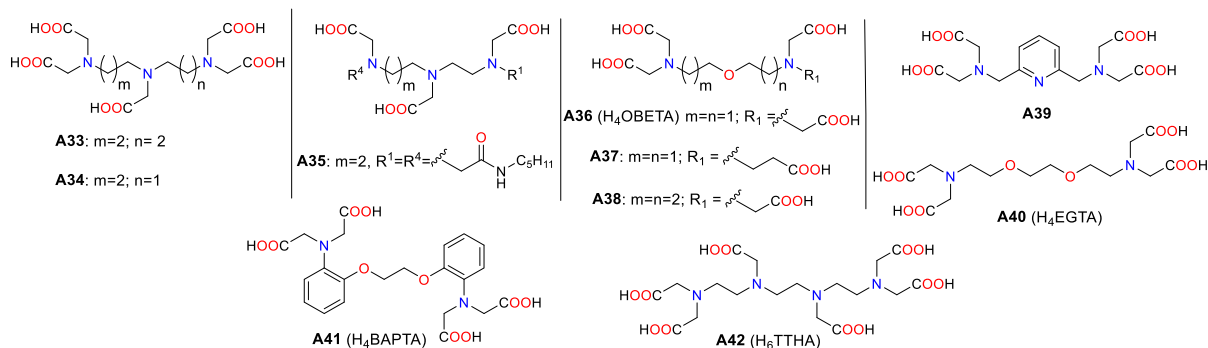
A third group of non-macrocyclic ligands is the family of tripodal ligands such as H₃NTA (**A62**) and H₂IDA (**A63**) [60,61], and H₂IDA derivatives containing alkyl (**A64**, **A66**) [25], hydroxyethyl (**A65**) [62], or phenol (**A67**) substituents [63]. The simplest ligand of this class considered in this work is glycine (**A68**) [64]. The most stable complex among this series of tripodal ligands is provided by **A67**, which suggest the favorable properties of the phenol group for stable Gd(III) complexation (Fig. 3). A class of tripodal ligands that has been rather intensively investigated is the HOPO family, such as TREN-1-Me-3,2-HOPO (**A72**) and TREN-6-Me-3,2-HOPO (**A73**) (Fig. 3). The Gd(III) complexes of these ligands and closely related derivatives were found to contain two or three water molecules coordinated to the metal ion, which results in remarkably high relaxation efficiencies [73,74]. The thermodynamic stability of the Gd(III) complexes of some HOPO derivatives was reported, though the amount of data available is scarce. The complexes of the parent HOPO ligands **A72** and **A73** are characterized by relatively high stability constants of log K_{GdL} = 19.2 and 19.5, respectively [66]. This stability is enhanced considerably by replacing one of the HOPO units by a terephthalamide moiety, with the complexes of **A74** and **A75** being characterized by log K_{GdL} = 24.3 and 24.1, respectively [66,69]. Other structural modifications introduced in one of the chains of the tripod, to give **A76-A78**, provoke a significant drop in complex stability [70].

Table 1

Stability constants reported in the literature for the Gd(III) complexes of linear ligands **A1–A78** and ionic strengths used for their determination.

	$\log K_{\text{GdL}}$	I	Ref		$\log K_{\text{GdL}}$	I	Ref
A1 (H ₅ DTPA)	22.03	0.15 M NaCl	[22]	A39	18.60	0.1 M KCl	[41]
	22.26	0.1 M NaCl	[23]	A40 (H ₄ EGTA)	17.66	0.1 M KCl	[42]
	22.46	0.1 M KNO ₃	[24]		17.50	0.1 M KNO ₃ [a]	[43]
A2	21.66	0.1 M Me ₄ NCl	[25]	A41 (H ₄ BAPTA)	10.6	0.1 M KNO ₃	[44]
A3	22.23	0.1 M Me ₄ NNO ₃	[25]	A42 (H ₆ TTHA)	28.4	0.1 M	[45]
A4	22.0	0.1 M KCl	[26]	A43 (H ₄ EDTA)	17.35	0.1 M	[46]
A5	18.60	0.1 M KCl	[27]	A44	12.79	0.1 M KCl	[47]
A6	17.79	0.1 M KNO ₃	[28]	A45	12.76	0.1 M KCl	[47]
A7	22.79	0.1 M KCl	[29]	A46	10.3	0.1 M Me ₄ NCl	[48]
A8	22.34	0.1 M Me ₄ NCl	[29]	A47	12.37	0.16 M NaCl	[49]
A9	19.90	0.1 M KCl	[30]	A48	8.1	0.1 M KNO ₃	[46]
A10	21.59	0.1 M Me ₄ NNO ₃	[31]	A49	18.89	0.1 M KCl	[50]
A11	21.72	0.1 M Me ₄ NNO ₃	[25]	A50 (H ₄ OCTAPA)	20.23	0.15 M NaCl	[51]
A12	16.30	0.1 M NaCl	[23]	A51	12.98	0.1 M KNO ₃	[52]
A13	13.12	0.1 M KCl	[37]	A52	12.40	0.1 M KNO ₃	[53]
A14	14.6	0.1 M KCl	[33]	A53	17.54	0.1 M KCl	[54]
A15 (H ₃ DTPA-BMEA)	16.84	0.1 M NaClO ₄	[34]	A54	15.56	0.1 M KCl	[55]
A16 (H ₃ DTPA-BMA)	16.72	0.15 M NaCl	[35]	A55	15.22	0.1 M KNO ₃	[56]
	16.85	0.1 M NaCl	[32]	A56 (H ₄ CDTA)	19.6	0.1 M KNO ₃	[46]
A17 (H ₃ DTPA-BPA):	16.23	0.1 M NaCl	[23]	A57 (H ₄ CHXOCTAPA)	20.68	0.15 M NaCl	[57]
A18	16.5	0.1 M KCl	[33]	A58 (H ₄ TMDTA)	13.76	0.1 M KNO ₃	[58]
A19	16.70	2 M NaClO ₄	[36]	A59	13.94	0.1 M KNO ₃	[58]
A20	16.17	2 M NaClO ₄	[36]	A60 (H ₄ PhDTA)	13.89	1 M NaClO ₄	[59]
A21	17.68	0.1 M NaClO ₄	[34]	A61	7.88	0.1 M	[25]
A22	17.49	0.1 M NaClO ₄	[34]	A62 (H ₃ NTA)	11.54	0.1 M KNO ₃	[60]
A23	19.0	0.1 M KCl	[33]	A63 (H ₂ IDA)	6.68	0.1 M KNO ₃	[61]
A24	16.83	0.1 M NaCl	[32]	A64	6.8	0.1 M	[25]
A25	19.68	0.1 M NaCl	[23]		6.68	0.1 M KCl	[25]
A26	18.91	0.1 M NaCl	[23]	A65	9.01	0.1 M KNO ₃	[62]
A27 (H ₅ BOPTA)	21.91	0.15 M NaCl	[22]	A66	7.04	0.1 M KCl	[25]
A28	19.74	0.1 M KNO ₃	[38]	A67	13.55	0.1 M KNO ₃	[63]
A29	19.4	0.1 M KCl	[30]	A68	3.72	0.1 M KNO ₃	[64]
A30	22.18	0.1 M Me ₄ NNO ₃	[25]	A69	14.73	0.15 M KCl	[65]
A31	22.06	0.1 M NaClO ₄	[39]	A70	17.41	0.15 M KCl	[65]
	23.2	0.1 M Me ₄ NCl	[39]	A71	18.00	0.15 M KCl	[65]
A32 (H ₅ EOB-DTPA)	23.46	0.1 M KCl	[40]	A72	19.2	0.1 M KCl	[66]
A33	13.00	0.1 M Me ₄ NCl	[67]	A73	20.3	0.1 M KCl	[66]
A34	17.5	0.1 M Me ₄ NCl	[67]	A74	24.3	0.1 M KCl	[66]
A35	13.44	0.1 M KCl	[68]	A75	24.1	0.1 M KCl	[69]
A36 (H ₄ OBETA)	19.37	0.1 M KCl	[42]	A76	14.8	0.1 M KCl	[70]
	18.21	0.1 M KNO ₃	[43]	A77	17.3	0.1 M KCl	[70]
A37	15.21	0.1 M KNO ₃	[71]	A78	16.5	0.1 M KCl	[70]
A38	11.74	0.1 M KNO ₃	[72]				

[a] Data at 20 °C.


Fig. 2. Structures of linear ligands structurally related to H₅DTPA.

As mentioned above, Gd(III) complexes for application as MRI contrast agents must present a high stability to prevent *in vivo* dissociation. Obviously many of the complexes listed in Table 1 do not meet this criterion. They were considered here because these stability data will aid correlating the stability constants with different structural descriptors, as will be discussed below.

2.2. Complexes with macrocyclic ligands

The family of macrocyclic ligands that has been more extensively investigated is the one derived from 1,4,7,10-tetraazacyclododecane (CYCLEN), a platform that has been functionalized with a wide variety of pendant arms. CYCLEN derivatives with four pendant arms are known to provide very

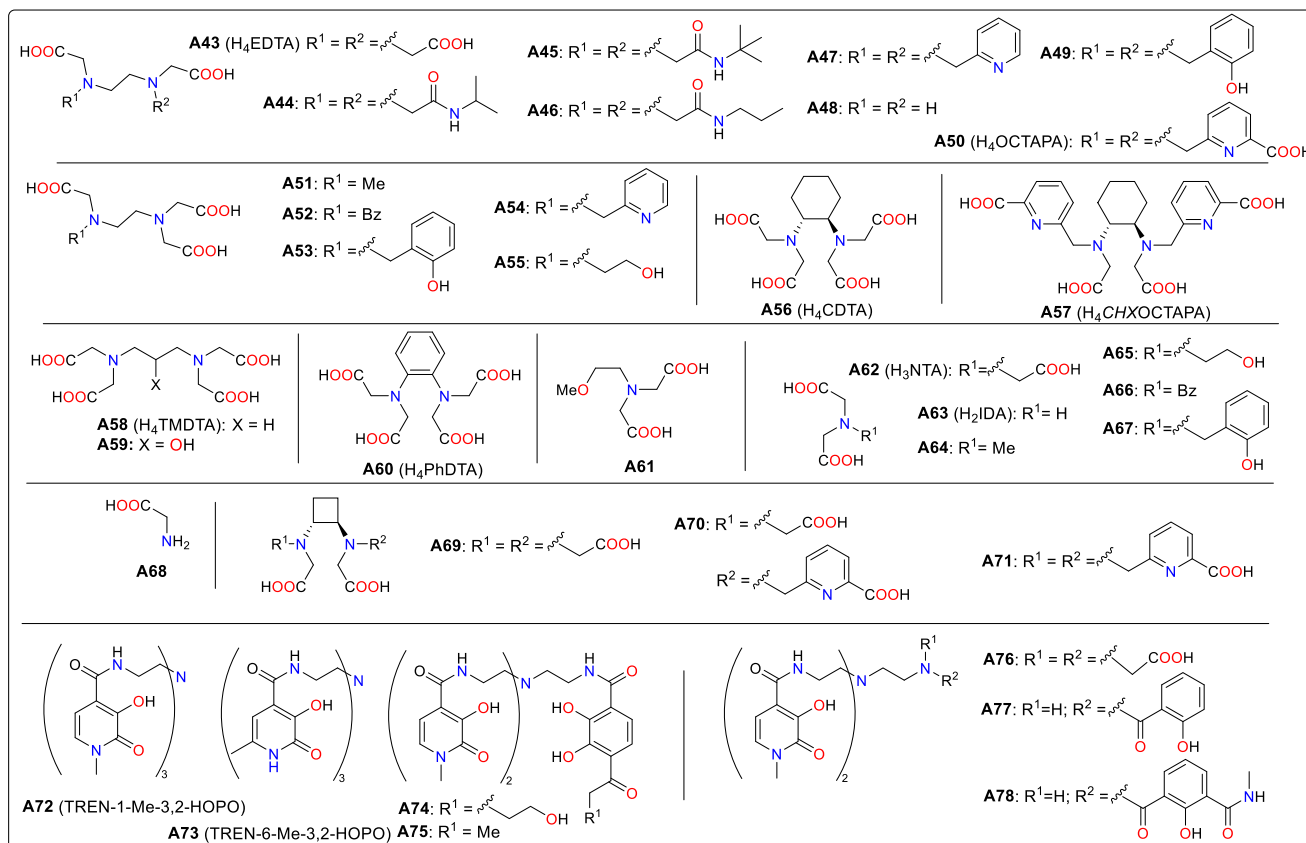


Fig. 3. Structures of H_4EDTA derivatives **A43**–**A60**, tripodal ligands **A61**–**A68** and cyclobutyl derivatives **A69**–**A71** and HOPO derivatives **A72**–**A78**.

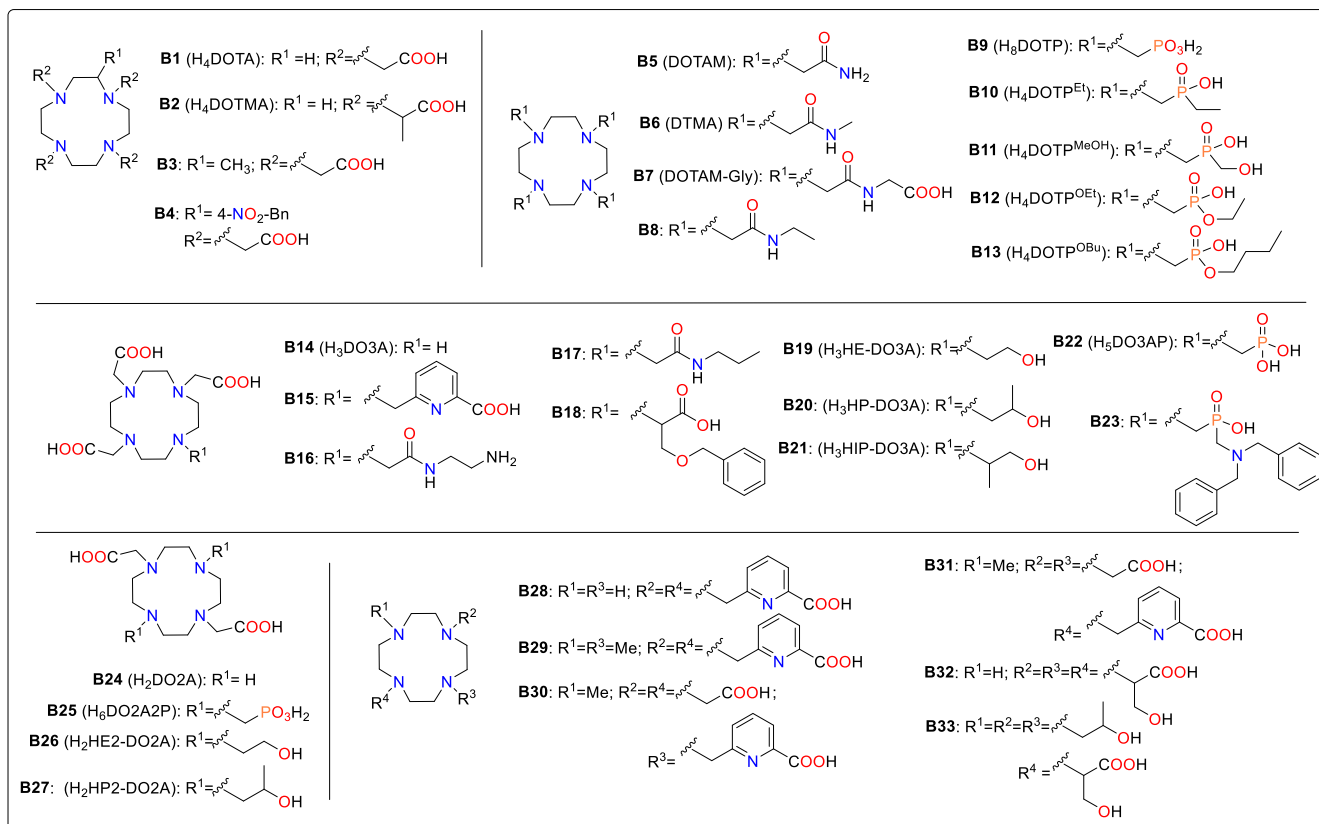


Fig. 4. Structures of CYCLEN-based ligands.

Table 2Stability constants reported in the literature for complexes with macrocyclic and mesocyclic ligands and ionic strengths used for their determination.^[a]

	log K_{GdL}	I	Ref		log K_{GdL}	I	Ref
B1 (H ₄ DOTA)	24.7	0.1 M KCl	[75]	B35	19.42	1 M KCl	[101]
B2 (H ₄ DOTMA)	23.6	0.1 M Me ₄ NNO ₃	[76]	B36 (H ₂ PC2A)	16.6	0.1 M KCl	[95]
B3	26.3	0.1 M	[82]	B37	22.37	0.15 M NaCl	[102]
B4	24.2	1 M Me ₄ NCl	[83]	B38	20.49	0.15 M NaCl	[102]
B5 (DOTAM)	13.12	1 M KCl	[77]	B39	23.44	0.15 M NaCl	[103]
B6 (DTMA)	13.54	1 M KCl	[77]	B40	23.56	0.15 M NaCl	[103]
B7 (DOTAM-Gly)	14.54	1 M KCl	[77]	B41	14.5	0.1 M KCl	[95]
B8	14.83	1 M KCl	[77]	B42	13.62	0.1 M KCl	[104]
B9 (H ₈ DOTP)	28.8	0.1 M Me ₄ NCl	[78]	B43	9.34	0.1 M ^[b]	[105]
B10 (H ₄ DOTP ^{Et})	16.50	0.1 M KCl	[79]	B44 (H ₄ TRITA)	19.17	0.1 M KCl	[106]
B11 (H ₄ DOTP ^{MeOH})	16.09	0.1 M	[80]	B45 (H ₄ TETA)	13.77	0.1 M KCl	[106]
B12 (H ₄ DOTP ^{OE})	14.4	0.1 M Me ₄ NCl	[80]	B46	18.82	0.1 M NaCl	[107]
B13 (H ₄ DOTP ^{OBu})	12.19	0.1 M Me ₄ NCl	[81]	B47 (H ₃ ODOTRA)	21.6	0.1 M KCl	[108]
B14 (H ₃ DO3A)	21.56	0.1 M KCl	[84]	C1 (H ₃ NOTA)	13.7	0.1 M KCl	[109]
	19.06	0.15 M NaCl	[85]	C2	13.9	0.1 M KCl	[110]
B15	23.31	0.1 M KCl	[88]	C3	15.8	0.1 M KCl	[111]
B16	22.25	0.1 M KCl	[86]	C4	15.10	0.1 M KCl	[112]
B17	20.1	0.1 M NaCl	[87]	C5	19.5	0.1 M KCl	[112]
B18	25.93	0.1 M Me ₄ NCl	[90]	D1 (H ₂ DACDA)	11.93	0.1 M Me ₄ NCl	[113]
B19	22.3	0.1 M Me ₄ NCl	[93]	D2 (H ₂ MACROPA)	13.02	0.1 M KCl	[114]
B20	23.8	0.1 M Me ₄ NCl	[93]	D3	7.02	0.1 M Me ₄ NCl	[115]
B21	23.9	0.1 M Me ₄ NCl	[93]	D4	7.29	0.1 M Me ₄ NCl	[115]
B22	27.5	0.1 M Me ₄ NCl	[91]	D5	18.02	0.1 M KCl	[108]
B23	23.77	0.1 M Me ₄ NCl	[92]	D6 (H ₄ PYTA)	21.7	0.1 M KCl	[116]
B24 (H ₂ DO2A)	19.1	0.1 M	[94]	E1	11.66	0.1 M Me ₄ NCl	[117]
B25 (H ₆ DO2A2P)	25.7	0.1 M KCl	[97]	E2	11.41	0.1 M KCl	[118]
B26 (H ₂ HE-DO2A)	21.1	0.1 M	[94]	E3	17.23	0.1 M KCl	[108]
B27 (H ₂ HP-DO2A)	22.5	0.1 M	[94]	E4	11.23	0.1 M NaClO ₄	[119]
B28	17.27	0.15 M NaCl	[98]	F1 (H ₄ AAZTA)	20.24	0.1 M KCl	[120]
B29	17.59	0.15 M NaCl	[98]		18.93	0.15 M NaCl	[121]
B30	16.69	0.15 M NaCl	[100]	F2	20.33	0.15 M NaCl	[121]
B31	17.12	0.15 M NaCl	[100]	F3	20.06	0.15 M NaCl	[121]
B32	18.82	0.1 M Me ₄ NCl	[99]	F4	18.26	0.1 M KCl	[122]
B33	19.4	0.1 M Me ₄ NCl	[99]	F5	25.06	0.15 M NaCl	[123]
B34 (H ₃ PCTA)	20.39	1 M KCl	[124]	F6	19.57	0.15 M NaCl	[123]
	18.28	0.15 M NaCl	[102]				

[a] All data at 25 °C. [b] 0.085 M KCl + 0.015Na⁺.

stable Gd(III) complexes that are often very inert with respect to their dissociation. The CYCLEN-based backbone functionalized with four acetic acid pendant arms (H₄DOTA, **B1**, Fig. 4) forms a very stable and inert Gd(III) complex that is widely used as a MRI contrast agent in clinical practice (gadoterate meglumine) [75]. The CYCLEN platform has been also functionalized with four identical pendant arms replacing the acetic acid arms of H₄DOTA. These pendant arms include 2-propionic acid (**B2**, H₄DOTMA) [76], acetamides (**B5–B8**) [77], methylphosphonic acid (**B9**, H₈-DOTP) [78], methylphosphonic acid (**B10** (H₄DOTP^{Et}); **B11** (H₄DOTP^{MeOH})) [79,80], and methylphosphonic acid esters (**B12** (H₄DOTP^{OE}), **B13** (H₄DOTP^{OBu})) [80,81]. Some closely related derivatives containing substituents on one carbon atom of the CYCLEN ring have been also investigated (**B3**, **B4**) [82,83]. The stability constants reported for these two closely related derivatives differ by 2.7 log K units, with that of [Gd(DOTA)(H₂O)][−] falling in between them (Table 2). The stability constants reported for the Gd(III) complexes of CYCLEN-based ligands containing four identical pendant arms vary dramatically depending on the nature of the side chains (Table 2). The most stable complex of this series is that with H₈DOTP (**B9**, log K_{GdL} = 28.8) [78], while its *n*-butyl ester derivative H₄DOTP^{OBu} presents a stability constant 16.6 log K units lower (**B13** (H₄DOTP^{OBu}), log K_{GdL} = 12.19) [81]. As observed for H₃-DTPA-bis(amide) derivatives, the CYCLEN-based ligand containing four acetamide pendants DOTAM (**B5**) forms a complex with a lower stability compared with the analogues containing substituted acetamides **B6–B8** [77].

Stability constants were also determined for complexes of several H₃DO3A derivatives. The stability constants of the complex

with H₃DO3A (**B14**, Fig. 4) determined using ionic strengths of 0.1 M KCl (log K_{GdL} = 21.56) and 0.15 M NaCl (log K_{GdL} = 19.06, Table 2) differ considerably [84,85]. A similar effect is observed comparing the stability constants determined for the complexes of the monoamide derivatives **B16** and **B17**. The stability constant of the complex with **B17** was determined in 0.1 M NaCl, and found to be ~2.2 log K units lower than that of **B16** determined in 0.1 M KCl [86,87]. These data indicate that these ligands form a rather stable complex with Na⁺ that competes with the Gd(III) ion, resulting in lower values of the stability constants. The stability constant of the Na⁺ complex of DOTA was estimated to be log K_{NaL} = 4.2 [88]. As a result, the basicity of CYCLEN derivatives varies considerably depending on the cation present in the background salt. For instance, the first protonation constant of DOTP^{8−} was determined to be log K_1 = 13.7, 12.1 and 10.9 in 0.1 M Me₄N(NO₃), KNO₃ and NaCl [89]. Thus, stability constants measured using different ionic strengths cannot be directly compared. This is in sharp contrast with the thermodynamic data reported for H₃DTPA derivatives, which do not differ significantly in different ionic strengths (i. e. log K_{GdL} values of 22.03, 22.26 and 22.46 were reported for GdDTPA^{2−} using I = 0.15 M NaCl, 0.1 M NaCl and 0.1 M KNO₃, respectively, Table 1).

In addition to acetamide groups, the H₃DO3A platform was also functionalized with picolinic acid (**B15**) [88], α -substituted acetic acid (**B18**) [90], methylphosphonic acid (**B22**) [91], or methylphosphonic acid groups (**B23**) [92]. The stability constants of H₃DO3A ligands containing hydroxyethyl (**B19**), 2-hydroxypropyl (**B20**), or 3-hydroxypropyl (**B21**) pendants suggest that the presence of the additional electron-donating methyl group in **B20** and **B21**

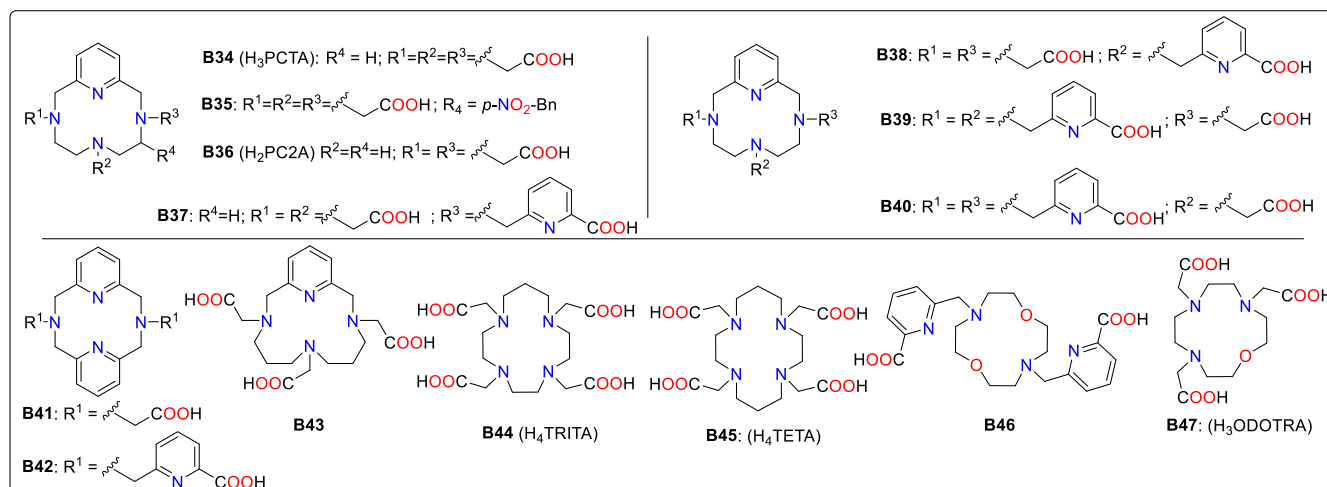


Fig. 5. Structures of macrocyclic ligands incorporating pyridyl rings, oxygen atoms or propyl chains into the macrocyclic structures.

increases the stability of the Gd(III) complexes by $\Delta \log K_{GdL} \sim 1.5$ with respect to the complex of **B19**, which contains an unsubstituted hydroxyethyl pendant [93].

The stability constants reported in the literature for the complex of H_2DO2A (**B24**) differ by up to 6 orders of magnitude. Two independent determinations yielded very similar stability constants ($\log K_{GdL} = 19.1$ and 19.42) [94,95], while a much lower value was determined by Chang ($\log K_{GdL} = 13.06$) [96]. These discrepancies are likely related to the slow kinetics of formation of this complex. A value of $\log K_{GdL}$ of about 19 appears to be more consistent with the stability decrease observed upon removing one of the acetic acid pendant arms of H_4DOTA ($\log K_{GdL} = 24.7$ for $[Gd(DOTA)(H_2O)]^-$) to give H_3DO3A ($\log K_{GdL} = 21.6$), which suggests that each carboxylate provides a contribution of about 3 $\log K$ units. Thus, one would expect a stability constant for the H_2DO2A derivative of $\log K_{GdL} \sim 18.6$, a value that further suggests that the stability constant reported by Chang is incorrect [96]. We thus retained in our analysis a stability constant of $\log K_{GdL} = 19.1$ (Table 2). The stabilities of the complexes with the H_2DO2A derivatives **B26** ($H_2HE\text{-}DO2A$) and **B27** ($H_2HP\text{-}DO2A$, Fig. 4, Table 2) confirm that ligands containing 2-hydroxypropyl groups form more stable complexes than the hydroxyethyl analogues (**B26**) [94].

The stability constant reported for the complex of $H_6DO2A2P$ (**B25**, Table 2) is intermediate between those of H_4DOTA and $H_8\text{-}DOTP$, suggesting a higher contribution to complex stability of the methylphosphonic acid compared with acetic acid arms [97]. A relatively large variety of asymmetrically-substituted CYCLEN derivatives, containing different types of pendant arms, were also used for Gd(III) complexation (Fig. 4, B28–B33) [98–100].

A structural modification of the CYCLEN platform that was investigated in detail is the incorporation of pyridine units to the macrocyclic scaffold (Fig. 5). The best known example of this class of ligands is the triacetic acid derivative H_3PCTA (**B34**, Fig. 5). As for CYCLEN derivatives, the stability constants of $[Gd(PCTA)(H_2O)_2]$ determined in 0.1 M KCl and 0.15 M NaCl differ considerably (more than 2 $\log K$ units) [102,124]. Other derivatives whose stability constants were determined are the bis(acetic) acid derivative H_2PC2A (**B36**) [95], as well as systems containing mixed acetic acid and picolinic acid pendant arms (**B37–B40**) [102,103]. These pendant arms were also incorporated to the 12-membered macrocycle containing two pyridyl units in the macrocyclic backbone (**B41–B42**) [95,104]. Additionally, structural modifications such as the incorporation of a C-alkyl group to H_3PCTA (**B35**) or replacing ethyl by propyl chains in the macrocyclic scaffold (**B43**) were also explored [101,105]. The same structural modification was

introduced to H_4DOTA , yielding the 13 and 14 membered macrocycles H_4TRITA (**B44**, Fig. 5) and H_4TETA (**B45**) [106]. The stabilities of complexes with 12-membered macrocycles in which amine N atoms were replaced by ether O atoms were also investigated (**B46** and **B47**, Fig. 5) [107,108].

The stability constants reported for $[Gd(PCTA)(H_2O)_2]$ ($H_3\text{-}PCTA = \text{B34}$) are ca. 1 $\log K$ unit lower than those of $[Gd(DO3A)(H_2O)_2]$ ($H_3DO3A = \text{B14}$, Table 2) determined using the same ionic strength, suggesting that the incorporation of a pyridine ring to the macrocyclic scaffold has a negative contribution to the stability constant (Table 2). However, the rigidifying effect introduced by the pyridyl ring was proved to enhance the kinetic inertness of the complex [124]. The comparison of the stability constants of the complexes with H_3DO3A (**B14**) and $H_3ODOTRA$ (**B47**) suggests that replacing an amine nitrogen atom by an ether oxygen atom has a minor impact in complex stability.

The only bidentate chelating unit that was incorporated to different macrocyclic platforms is the 6-methylpicolinic acid group. In some instances the presence of the picolinic acid group provided an additional stability to the Gd(III) complex compared to the corresponding derivative containing acetic acid pendant arms. This is for instance the case of the complexes with **B37** and **B38**, which present $\log K_{GdL}$ values of 22.47 and 20.39 ($I = 0.15$ M NaCl) [102]. These stability constants are clearly higher than that determined for the H_3PCTA (**B34**) complex under the same conditions ($\log K_{GdL} = 18.28$) [102]. However, it is also noteworthy that the complexes of **B37** and **B38** present stability constants that differ by 2 $\log K$ units in spite of having identical donor sets. This indicates that steric effects are likely playing a role in decreasing the stability of the complex with **B38**. Replacing an additional acetic acid group of **B37** and **B38** by a picolinic acid pendant arm further increases the stability of the corresponding Gd(III) complexes ($\log K_{GdL} = 23.44$ and 23.56 for $Gd\text{B39}$ and $Gd\text{B40}$, respectively) [103]. The steric hindrance associated to the coordination of the bidentate picolinate unit is particularly evident for CYCLEN-based systems. Indeed, the stability constant determined for the complex of **B15** ($\log K_{GdL} = 23.31$) [88], which contains a picolinate arm replacing one of the acetate groups of $[Gd(DOTA)(H_2O)]^-$, is clearly lower than that of the latter ($\log K_{GdL} = 24.7$) [75]. Additional examples are provided by complexes $Gd\text{B30}$ and $Gd\text{B31}$, which present stability constants lower than that of the parent $[Gd(DO3A)(H_2O)_2]$ derivative ($Gd\text{B14}$) recorded under the same conditions (Table 2) [85,100].

The number of stability constants reported for Gd(III) complexes with macrocyclic ligands based on 9, 15 or 18 membered

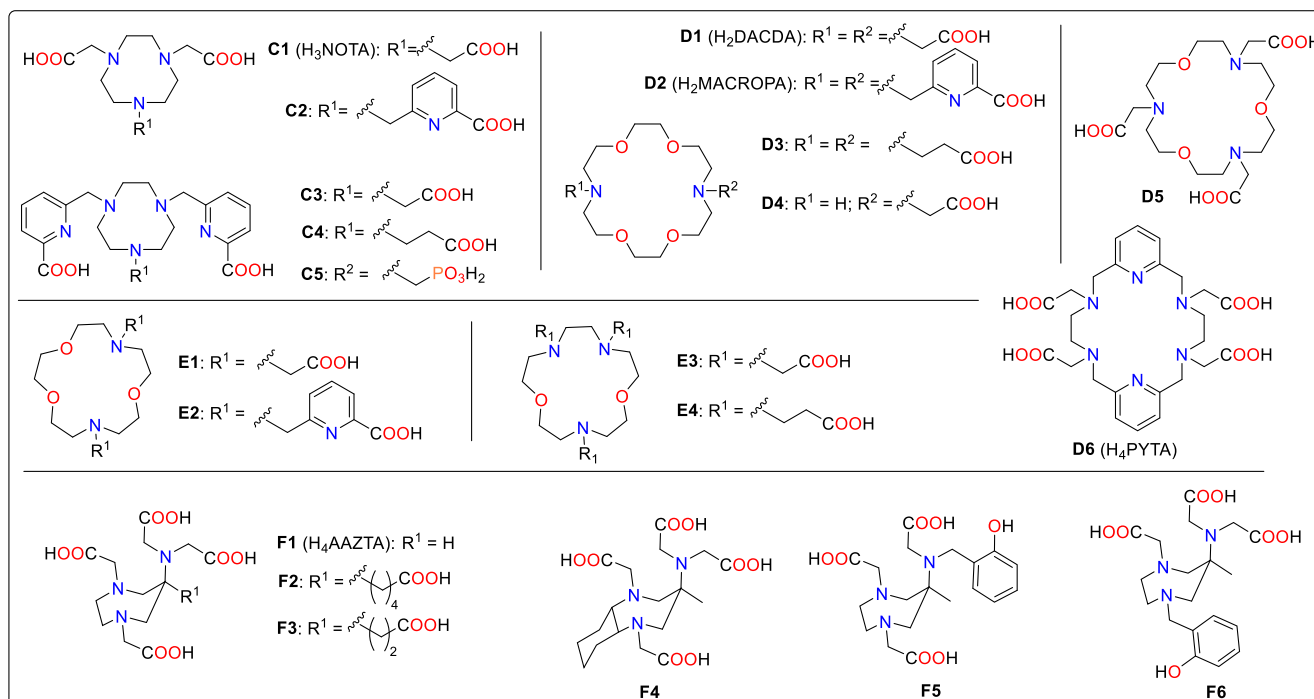


Fig. 6. Structures of macrocyclic ligands based on 9-, 15- and 18-membered macrocycles and mesocyclic H_4AAZTA derivatives.

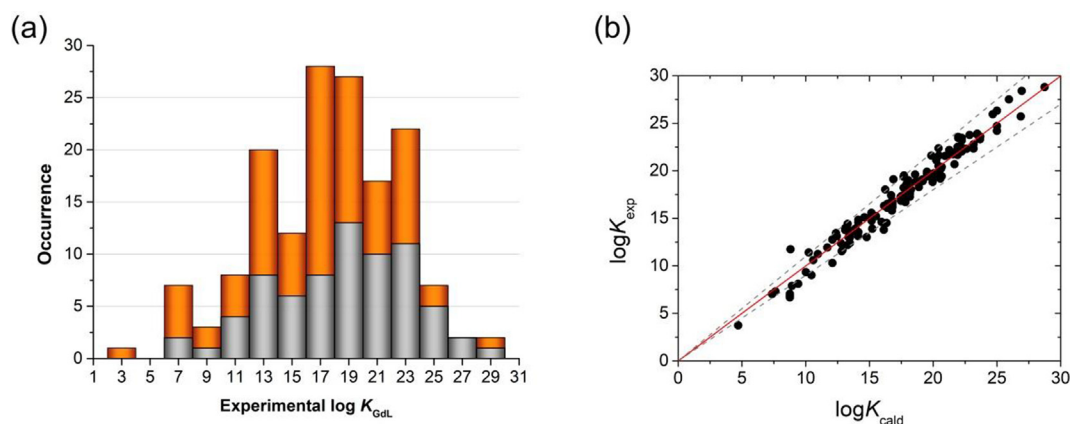


Fig. 7. (a) Histogram of the $\log K_{GdL}$ values for 156 data points (146 complexes) of macrocyclic and mesocyclic (gray) and acyclic (orange) systems and (b) Plot of the experimental stability constants (142 values) versus those calculated with Equation [1]. The red line is the line of identity and the dashed lines represent a $\pm 10\%$ deviation from a slope of unity.

macrocycles is unfortunately quite scarce. A few triazacyclononane (TACN) derivatives containing acetic acid, picolinic acid, propionic acid or methylphosphonic acid pendant arms have been described and the stability constants of the $Gd(III)$ complexes determined (**C1–C5**, Fig. 6) [109–112]. The family of 18-membered macrocycles includes H_2DACDA (**D1**) and the corresponding picolinic acid derivative $H_2MACROPA$ (**D2**), which was shown to be very selective for large metal ions [113,114]. The analogue of H_2DACDA containing propionic acid pendant arms (**D3**) and the monocarboxylic acid derivative **D4** were also investigated [115]. Besides these 1,10-diaza-18-crown-6 derivatives, stability data were also reported for the triacetic acid derivative of 1,7,13-triaza-18-crown-6 **D5** and the tetraacetic acid derivative of a 18-membered macrocycle containing pyridyl head units H_4PYTA (**D6**) [108,116,108,116]. Concerning the stability of complexes with 15-membered macrocycles, only four examples have been investigated (**E1–E4**, Fig. 6) [108,117–119].

A class of ligands that was developed specifically for formation of stable $Gd(III)$ complexes is the H_4AAZTA family, which are mesocyclic ligands based on a 6-amino-6-methylperhydro-1,4-diazepine platform (Fig. 6) [125]. While these ligands are not macrocyclic, strictly speaking, they show properties that are closer to macrocyclic systems, and thus are discussed here. The thermodynamic data available in the literature for $Gd(III)$ –AAZTA derivatives is scarce, with only a few stability constants reported. The stability of the $Gd(III)$ complex with H_4AAZTA (**F1**, Fig. 6) determined using 0.1 M KCl and 0.15 M NaCl backgrounds differ by $\sim 1.3 \log K$ units (Table 2), a behavior that parallels CYCLEN derivatives and implies affinity for the Na^+ ion. The presence of non-coordinating C-alkyl groups in **F2** and **F3** causes a slight increase in complex stability, which was attributed to an increased ligand basicity [121]. Rigidification of the H_4AAZTA structure with a cyclohexyl ring resulted in a decrease in the stability constant determined using the same ionic strength ($\sim 2 \log K$ units) [122].

A recent study reported the stability constants of the Gd(III) complexes with the regioisomeric ligands **F5** and **F6**, which contain a phenol group either at the exocyclic N atom of the H₄AAZTA scaffold (**F5**) or on one of the endocyclic N atoms (**F6**). The complex with **F5** is characterized by a particularly high stability constant ($\log K_{\text{GdL}} = 25.06$), which is 5.5 log K units higher than that of the **F6** derivative. This astonishing difference was explained by steric effects that hinder the coordination of the bulky phenol group attached to an endocyclic N atom [123].

3. Analysis of complex stability using structural descriptors

3.1. Estimation of stability constants

The $\log K_{\text{GdL}}$ values described in the previous section cover a wide range of values from 3.7 for the glycinate complex to 28.8 for the complex of H₈DOTP (Fig. 7). For 85% of the macrocyclic systems the stability constant values (60) fall within the range $10 < \log K_{\text{GdL}} < 24$. Up to 76% (74) of the $\log K_{\text{GdL}}$ values of complexes with acyclic ligands are within the $10 < \log K_{\text{GdL}} < 24$ range, with only three values above this range (H₆TTHA (**A42**) and the HOPO derivatives **A74** and **A75**, Table 1).

Given the rather large number of Gd(III) complexes whose stability was investigated, we postulated that these data could be used to obtain an empirical expression allowing to predict and rationalize the stability constants. We hypothesized that the stability constants could be approximated by an expression of the form:

$$\log K_{\text{GdL}} = \sum_i n_i \Delta \log K_i + \sum_j n_j \Delta \log K_j + A_{\text{GdL}} \quad [1]$$

In the first summation, n_i represents the number of donor groups of type i and $\Delta \log K_i$ the contribution of this type of donor group to the stability constant. The second summation describes the effect of different structural modifications, with n_j being the number of such modifications and $\Delta \log K_j$ its effect on complex stability. For the complexes with acyclic ligands A_{GdL} was set to zero. The different donor groups of the ligand are denoted as N (amine nitrogen atoms), C (carboxylate groups), C α (α -substituted carboxylate), C_{es} (carboxylate ester), A_{NH2} (primary amide), A_{NHR} (secondary amide), A_{NR2} (tertiary amide), Pho (phosphonate), Pic (picolinate), Prop (propionate), Py (pyridine), Phe (phenol), HE (hydroxyethyl), HP (hydroxypropyl), Phosphi (phosphinate) and Phoses (phosphonate ester). Thus, H₄EDTA (**A43**) is characterized by $n_N = 2 + n_C = 4$, while H₅DTPA (**A1**) is described by $n_N = 3 + n_C = 5$ and H₃DTPA-BMA (**A19**) as $n_N = 3 + n_C = 3 + n_{\text{ANHR}} = 2$. The structural modifications considered are replacing an ethylene chain by a propylene (SPropyl), cycloheptyl (SCyhx), cyclobutyl (SCybu) or phenyl (SPh) group, and substituting an amine nitrogen atom by an ether oxygen atom (SOe) or a pyridyl ring (SPy). For instance, H₄CDTA (**A56**, Fig. 3) is described by $n_N = 2 + n_C = 4 + n_{\text{SCyhx}} = 1$, while H₄EGTA is defined with $n_N = 4 + n_C = 4 + n_{\text{SOe}} = 2$ (two ether oxygen atoms are replacing two of the four amine nitrogen atoms). Note that while the $\Delta \log K_i$ terms must be positive, since all donor groups should have a certain contribution to complex stability, the $\Delta \log K_j$ values might be positive or negative depending on the effect of such structural modification. Inspection of the stability constants shows that C-functionalization of the ligand scaffold does not affect significantly the stability constants, and thus this structural modification was not considered.

For the description of the complexes with macrocyclic ligands we had to use a slightly different approach to account for the intrinsic coordination properties of the macrocycle, in terms not only of number (and type) of donor atoms, but also the size match between the macrocycle and the metal ion. Thus, for macrocyclic

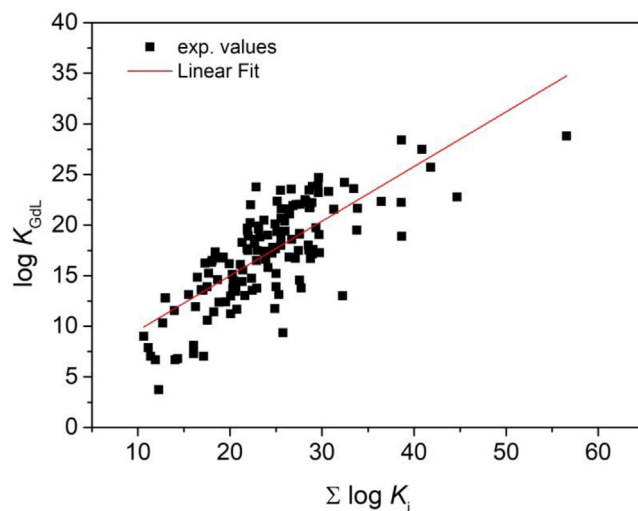


Fig. 8. Plot of $\log K_{\text{GdL}}$ vs the sum of the logarithmic values of the ligand protonation constants for 133 Gd complexes.

ligands we used a term A_{GdL} to capture the contribution to the stability constant of the macrocyclic unit, while the first summation described the contributions of the donor groups of the pendant arms only, and the second summation the structural modifications of the macrocyclic ring and pendant arms.

Following the qualitative observations described above, we have used two different $\Delta \log K_{\text{Pic}}$ values as fitting parameters, to account for sterically hindered ($\Delta \log K_{\text{Pic}}^*$) and non-hindered picolinate groups. The criterion followed to choose one or another parameter was the comparison of the stability constants of the carboxylate and picolinate analogues. A sterically hindered picolinate was considered in those cases in which the stability of the carboxylate derivative is higher than that of the picolinate analogue (i. e. **B1** and **B15**). Concerning the effect of the ionic strength, in the case of complexes with macrocyclic ligands derived from CYCLEN and structural modifications, we used two different parameters A as fitting parameters to account for ionic strengths adjusted with NaCl or other electrolyte.

An expression such as that defined in [1] yields higher calculated stability constants on increasing the denticity of the ligand. However, the Gd(III) ion rarely exceeds coordination number nine in aqueous solution, reaching coordination number ten in some cases [126]. All ligands described in the previous section are nonadentate with the exceptions of **A42** (H₆TTHA), **D2** (H₂MACROPA) and **D6** (H₄PYTA). The crystal structures of TTHA⁶⁻ complexes with the larger Ln(III) ions show the metal ion coordinated in a decadenate fashion, while complexes of the smaller Ln(III) ions show 9-fold coordination [127,128]. The Gd(III) complexes of **D2** (H₂MACROPA) and **D6** (H₄PYTA) were shown to present ten-coordinate metal ions both in the solid state and in solution [114,129].

A least squares fit of the stability constants reported for the 133 complexes (142 stability constant values) with the ligands shown in Figs. 1–6 provided calculated stability constants in excellent agreement with the experimental values, as evidenced in Fig. 7 ($R^2 = 0.997$). The least-squares fit involved 26 fitting parameters: The $\Delta \log K_i$ of 16 donor groups, 5 $\Delta \log K_j$ parameters describing the different structural modifications and 5 A_{GdL} parameters (1 for the 9, 15 and 18 membered macrocycles and 2 for the 12-membered macrocycles). The average mean deviation of the calculated values with respect to the experimental ones is 0.75 log K units, with the largest deviations being observed for the complexes of H₄BAPTA (**A41**), H₂DO2A (**B24**), HP-DO3A-3HM (**B33**) and

Table 3Parameters obtained from the least-squares fit of the stability data according to equation [1] and number of entries of each structural feature ($\sum n_{ij}$).^[a]

	$\Delta \log K_{ij}$	Sn_{ij}	ΔpGd_{ij}	Sn_{ij}
N	0.65(0.32)	221	-0.35(0.36)	180
C	4.06(0.20)	405	4.00(0.22)	338
C $_{\alpha}$	3.73(0.26)	18	3.93(0.28)	18
C $_{es}$	1.02(0.51)	3	2.12(0.53)	3
A $_{NH2}$	0.92(0.30)	6	2.04(0.32)	6
A $_{NHR}$	1.34(0.21)	43	2.47(0.23)	37
A $_{NR2}$	1.97(0.36)	6	2.95(0.38)	6
Pho	5.00(0.29)	11	4.11(0.30)	10
Pic	5.60(0.28)	31	6.17(0.19)	28
Pic*	2.73(0.41)	7	3.99(0.41)	7
Prop	1.89(0.35)	8	2.62(0.37)	8
Py	1.68(0.37)	6	2.53(0.39)	6
Phe	4.63(0.44)	4	1.37(0.45)	4
HE	1.68(0.43)	5	2.67(0.44)	5
HP	2.52(0.34)	7	2.89(0.35)	7
Phosphi	1.91(0.26)	9	2.44(0.28)	9
Phoses	1.13(0.26)	8	1.74(0.38)	8
Spropyl	-4.44(0.25)	13	-4.67(0.26)	13
SCyhx	1.06(0.80)	2	0.14(0.82)	2
SCybu	-2.36(0.67)	3	-1.80(0.69)	3
SPh	-3.62(0.50)	3	-1.84(0.52)	3
SOe	-0.49(0.29)	43	0.62(0.32)	40
SPy	-0.27(0.32)	16	1.39(0.34)	15
A $_9$	1.50(0.80)	5	-2.47(0.83)	5
A $_{12}$	8.76(0.75)	36	3.41(0.83)	35
A $_{12}$ (NaCl)	6.97(0.74)	12	3.12(0.83)	12
A $_{15}$	6.25(1.26)	4	1.11(1.40)	4
A $_{18}$	5.55(1.40)	6	0.06(1.53)	6

^[a] Standard deviations are provided within parentheses. N, (amine nitrogen atoms); C, carboxylate groups; C $_{\alpha}$, α -substituted carboxylate; C $_{es}$, carboxylate ester; A $_{NH2}$, primary amide; A $_{NHR}$, secondary amide; A $_{NR2}$, tertiary amide; Pho, phosphonate; Pic, picolinate; Prop (propionate), Py (pyridine), Phe (phenol), HE (hydroxyethyl), HP (hydroxypropyl), Phosphi (phosphinate) and Phoses (phosphonate ester); SPropyl, ethylene replaced by a propylene; SCyhx, ethylene replaced by cyclohexyl; SCybu, ethylene replaced by cyclobutyl; SPh, ethylene replaced by Phenyl; SOe, amine N atom replaced by ether O atom; SPy, amine N atom replaced by a pyridyl ring.

H $_4$ TETA (**B45**) (with deviations of 2.88, 2.25, 2.54 and 2.26 log K units, respectively). Among the 142 data points, only 18 of them have log K values deviating more than 10% from the measured value, a situation that affects more frequently complexes with low stability constants. The value of log K_{GdL} calculated for the DO2A complex (16.85) is somewhat lower than the experimental value of 19.1 reported by Sherry [94], but confirms that the stability constant determined by Chang (log K_{GdL} = 13.06) is likely wrong [96]. The high stability constant of H $_2$ DO2A is likely related to a lower steric hindrance associated to the coordination of the ligand, as the acetic acid arms occupy opposite positions in the CYCLEN framework. The ligands of the HOPO and AAZTA families were not included in the analysis because the limited number of stability constants reported in the literature prevented the estimation of robust structural descriptors.

The accuracy of the predicted stability constants is remarkable, considering: 1) The errors associated to the determination of the experimental stability constants and the effect of using different ionic strengths. For instance, log K_{GdL} values differing by up to 2.2 log K units (in the range 20.7 to 23.0, 25 °C) were reported for [Gd(DTPA)(H $_2$ O)] $^{2-}$ using ionic strengths adjusted to 0.1 M with different electrolytes [130]. The discrepancies are even more obvious for complexes with macrocyclic ligands, which might be related to a more important effect of the electrolytes used in the determination and the slow complexation kinetics often observed for complexes with macrocyclic ligands. This is clearly illustrated by the stability constants reported for [Gd(DOTA)(H $_2$ O)] $^{-}$, which differ by up to 5.9 log K units [130]; 2) Our approach to calculate the stability constants is based on structural descriptors, which consider only the number and type of donor groups and a series of structural modifications that can be introduced in the ligand

scaffold. However, the arrangement of the donor atoms of the ligand around the metal ion may also have an important impact in complex stability as a result of steric interactions [100]. This effect is clearly observed when comparing the stabilities of the complexes with **B37** and **B38**, which contain the same number and type of pendant groups arranged in a different manner on the macrocyclic platform. However, the corresponding stability constants (log K_{GdL} = 22.37 and 20.49, 0.15 M NaCl, 25 °C) differ by 1.9 log K units [102].

The parameters obtained from the least-squares fit of the stability data are shown in Table 3, together with their corresponding standard deviations estimated from the least-squares fitting. The largest standard deviations are generally observed for those structural descriptors with lower number of entries among the collection of complexes considered (lower $\sum n_{ij}$ values). Among the different monodentate binding motifs, the negatively charged phosphonate ($\Delta \log K_{pho}$ = 5.00), phenolate ($\Delta \log K_{phe}$ = 4.63) and carboxylate ($\Delta \log K_C$ = 4.06) groups provide the highest contributions to complex stability. An α -substituted carboxylate ($\Delta \log K_{C\alpha}$ = 3.73) yields a slightly lower contribution than the unsubstituted carboxylate. Even lower is the contribution to complex stability of a propionate pendant arm ($\Delta \log K_{prop}$ = 1.89), which is certainly associated to the formation of a less-favorable six-membered chelate ring. All remaining (charge-neutral) monodentate binding groups provide rather low contributions in the following order: $\Delta \log K_{HP} > \Delta \log K_{ANR2} \sim \Delta \log K_{phosphi} > \Delta \log K_{HE} \sim \Delta \log K_{Py} > \Delta \log K_{ANHR} > \Delta \log K_{phosest} > \Delta \log K_{Ces} > \Delta \log K_{ANH2} > \Delta \log K_N$. Concerning the bidentate picolinate unit, its contribution to complex stability appears to be quite variable depending on the particular ligand architecture, with values of $\Delta \log K_{pic}$ = 5.60 and $\Delta \log K_{pic^*}$ = 2.73 for a non hindered and a sterically hindered picolinate,

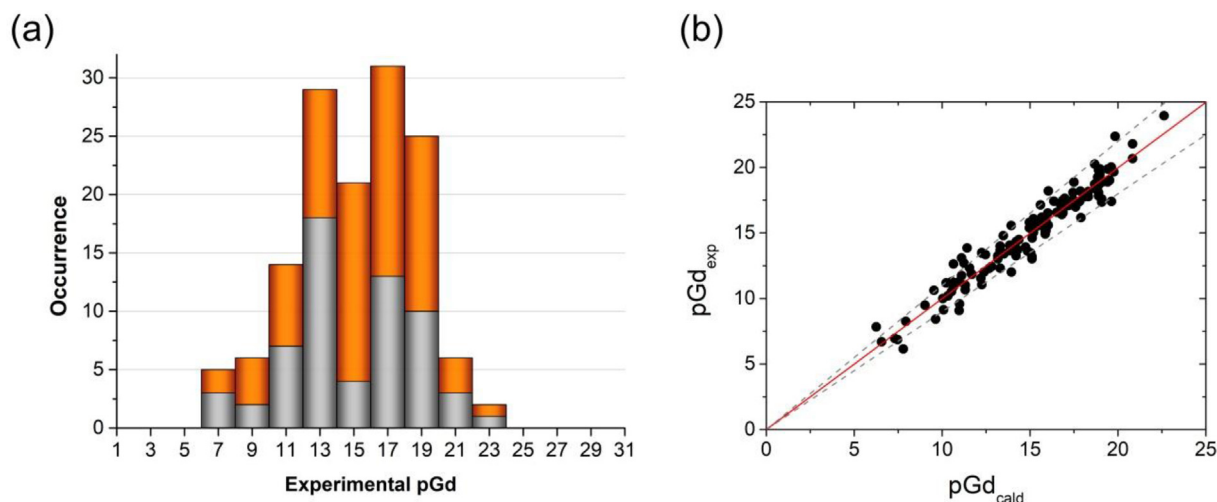


Fig. 9. (a) Histogram of the pGd values for 132 values of macrocyclic and mesocyclic (gray) and acyclic (orange) systems and (b) Plot of the experimental pGd values (132 values) versus those calculated with Equation [2]. The red line is the line of identity and the dashed lines represent a $\pm 10\%$ deviation from a slope of unity.

respectively. The contribution of a non-hindered picolinate is roughly the sum of the contributions provided by a carboxylate and a pyridine donor (5.74).

The A_{GdL} values obtained from the analysis of the data indicate that the tridentate unit of TACN (A_9) gives a very small contribution to complex stability, likely reflecting the poor match between the size of the cavity and that of the Gd(III) ion. The largest value of A is calculated for the 12-membered macrocycle CYCLEN, and then drops as the size of the macrocyclic cavity increases to 15 and 18 members (Table 3).

3.2. Estimation of pGd

The values of $\log K_{GdL}$ do not take into account the basicity of the ligand, as an increased ligand basicity generally increases the $\log K$ values, but also reduces the stability constant of the complex at physiological pH due to a more efficient proton competition. Indeed, different authors observed linear relationships between the $\log K_{GdL}$ values and the overall ligand basicity expressed as the sum of the ligand protonation constants $\sum \log K_i$, generally considering only structurally related ligands [94,131]. The large set of data analyzed here display a relatively weak linear correlation ($R^2 = 0.59$), as demonstrated by the Pearson's correlation coefficient of 0.77 (Fig. 8). Nevertheless, it is evident that ligand basicity must be taken into account to assess complex stability under physiologically relevant conditions.

One method to take into account the competition with acid is to calculate the conditional stability constant at a given pH value, e.g. pH 7.4. A related method is to calculate the pGd value at a given pH under standardized conditions, defined here as $-\log[Gd(III)]_{free}$ at pH 7.4 for a total Gd(III) concentration of 10^{-6} M and a ligand concentration of 10^{-5} M [132]. This is the most common definition used in the literature, though it is evident that MRI contrast agents are thermodynamically very stable and are not injected using a 10-fold ligand excess. Thus, these pGd values should not be considered as a measure of the actual free Gd(III) present *in vivo* if thermodynamic equilibrium was reached, but rather a tool to compare the stabilities of complexes with ligands having different basicities. The lowest value of pGd according to the definition provided above is $pGd = 6$, a situation in which all Gd(III) is in the uncomplexed form at pH 7.4. The higher the pGd value, the more stable the complex at pH 7.4. The values of pGd cover a narrower range of $6 < pGd < 24$ compared to the $\log K_{GdL}$ values (Fig. 9). Noteworthy

is the fact that complexes with macrocyclic and acyclic ligands show similar distributions, with most complexes falling in the range $11 < pGd < 20$, and a few examples with $pGd > 19$. Thus, the higher $\log K_{GdL}$ values often observed for macrocyclic complexes are not necessarily reflected in the corresponding pGd values, an effect that can be explained by the high basicity of many macrocyclic ligands, particularly those derived from CYCLEN.

Given that Eq [1] predicts the stability constants of Gd(III) complexes to a rather good accuracy, we next assessed whether a similar expression can be used to predict pGd values. Thus, we performed a least squares fit of the available pGd values according to the following expression:

$$pGd = \sum_i n_i \Delta pGd_i + \sum_j n_j \Delta pGd_j + A_{pGd} \quad [2]$$

Here the fitting parameters ΔpGd_i and ΔpGd_j represent the contributions to pGd of the different donor groups and structural modifications, respectively, and A_{pGd} is a parameter that characterizes the contribution to pGd of the different macrocyclic platforms depending on their size. The fit of the data included 132 pGd values, a number that is somewhat lower than the one used to analyse $\log K_{GdL}$ data. This is explained by the lack of reported ligand protonation constants determined under the same conditions as the $\log K_{GdL}$ values. Equation [2] gives calculated pGd values in reasonably good agreement with those obtained from the experimental equilibrium constants ($R^2 = 0.996$, Fig. 9), though the agreement is not as good as for $\log K_{GdL}$ data. The largest deviations (~ 2.5 pGd units) are observed for H_2DO_2A (B24) and H_8DOTP (B9), with only five values presenting deviations > 2.0 pGd units, and an average absolute deviation of 0.72 pGd units. The particularly large deviation observed for the $\log K_{GdL}$ and pGd values of H_2DO_2A suggests that the experimental data must be treated with some caution and probably reassessed.

4. Structural factors influencing stability at pH 7.4

The comparison of the parameters obtained for the different structural descriptors through Eqs. [1] and [2] yields important stability information. For instance, for most neutral donor atoms ΔpGd_i values are higher than the corresponding $\Delta \log K_i$ data, an effect that can be attributed to the low basicity of these donor groups (Fig. 10). That is, these neutral donors contribute to complex stability at pH 7.4 in a greater manner than predicted by their

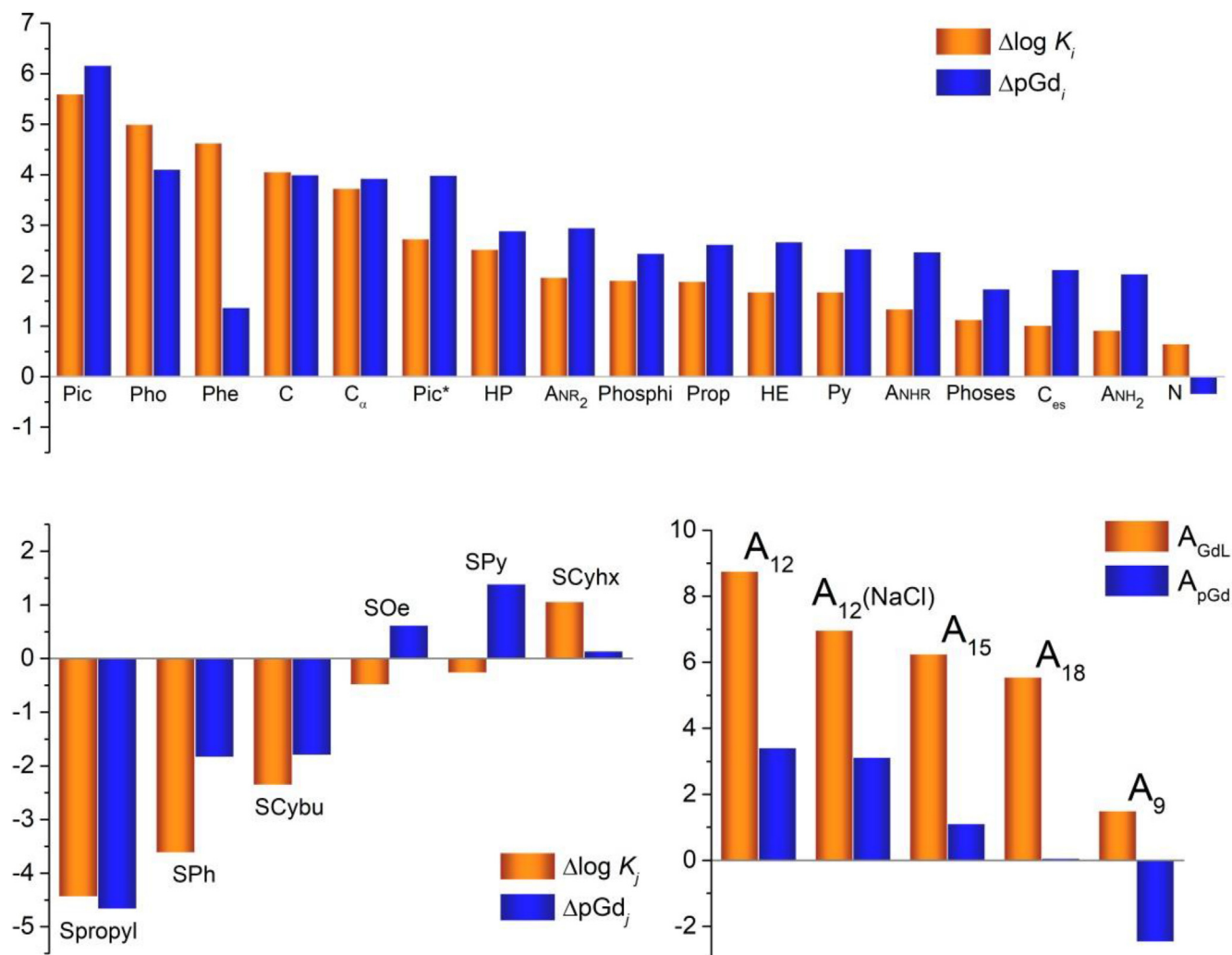


Fig. 10. Comparison of parameters $\Delta \log K_i$, ΔpGd_i , A_{GdL} and A_{pGd} obtained from the fits of the experimental $\log K_{GdL}$ and pGd values according to expressions [1] and [2].

basicity. The exception is observed for amine N atoms, which are obviously rather basic. As a result, each amine N atom provides a positive contribution to $\log K$ ($\Delta \log K_i = 0.65$), but a negative contribution to pGd ($\Delta pGd_i = -0.35$, Table 3). Carboxylate donor groups present very similar values of $\Delta \log K_i$ and pGd , while the bidentate picolinate groups display higher pGd values, an effect that is likely related to the lower overall basicity of ligands containing picolinic acid groups with respect to the carboxylic acid analogues [133]. For instance, the protonation of the amine N atoms of H₄EDTA (**A43**) are characterized by higher protonation constants ($\log K_1^H = 9.18$ and $\log K_2^H = 6.00$) than for H₄OCTAPA (**A50**, $\log K_1^H = 8.52$ and $\log K_2^H = 5.40$) [51]. A similar situation ($\Delta pGd_i > \Delta \log K_i$) is also observed for the propionate group, and can be attributed to the lower basicity of the amine N atom functionalized with a propionate group compared with the acetic acid analogue (i. e. the $\log K_1^H = 10.58$ and 9.64 for **A1** (H₅DTPA) and **A28**, respectively).

The value of ΔpGd_i calculated for phosphonate groups is only ~ 0.9 units lower than the corresponding $\Delta \log K_i$ parameter, which confers to this group the highest contribution to complex stability among the monodentate binding units, followed closely by the carboxylate group. Noteworthy is the much lower value of ΔpGd_i obtained for the phenol group ($\Delta pGd_i = 1.37$) compared with $\Delta \log K_i$ (4.63). This is clearly related to the high basicities of this group, as illustrated by the pK_a of o-cresol (10.26) [134]. Thus, while phe-

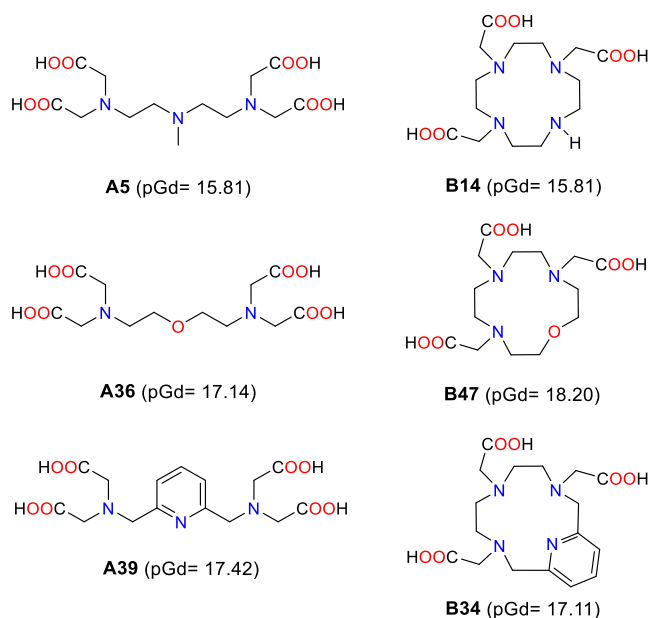


Fig. 11. Examples of structural modifications leading to increased pGd values.

nol groups give a very high contribution to the equilibrium constant, higher than that of carboxylates for instance, their high basicity results in a rather low contribution to stability at pH 7.4.

Concerning the values of $\Delta \log K_j$ and ΔpGd_j , replacing an ethylenediamine spacer by a propylenediamine group is clearly detrimental for both $\log K$ and pGd , as judged by the very negative $\Delta \log K_j$ and ΔpGd_j contributions. This is in line with earlier observations reported by Hancock, which demonstrated that large metal ions, like Gd(III), prefer five-membered chelate rings, while small metal ions form more stable six-membered chelate rings [135]. The incorporation of a phenyl spacer is characterized by a very negative $\Delta \log K_j$ value (−3.62) and also a negative ΔpGd_j contribution (−1.84). The lower destabilizing contribution observed for pGd is again related to the lower basicity of anilines compared to aliphatic amines [136]. Similarly, the low basicity of ether oxygen atoms and pyridyl N atoms with respect to amine N atoms provokes a change from slightly negative $\Delta \log K_j$ values to positive ΔpGd_j parameters. Thus, while replacing amine N atoms by ether or pyridyl groups generally has a minor impact in terms of stability constants, it usually results in increased pGd values. This is illustrated in Fig. 11, which presents the pGd values determined for a series of structurally related ligands in which amine N atoms are replaced by ether or pyridyl groups. The fact that $\Delta pGd_j < \Delta \log K_j$ for a cyclohexyl substituent also correlates with the higher basicity of ligands incorporating a cyclohexyl group instead of an ethyl spacer (i. e. **A50**, $H_4OCTAPA$, and **A57**, $H_4CHXOCTAPA$) [57]. Concerning the A_{pGd} values, 12-, 15- and 18-membered macrocycles are characterized by positive contributions, while 9-membered macrocycles are characterized by negative values of A_{pGd} . Interestingly, the Gd(III) complex of the 18-membered macrocycle **D6** (H_4PYTA) is characterized by a high stability constant ($\log K_{GdL} = 21.7$) and a high pGd value. However, structural modifications of this platform aimed to increase complex stability were not developed so far. Our model suggests that stabilities as high as $\log K_{GdL} = 25.0$ and a pGd value of 19.3 could be attained with the tetraphosphonate derivative, if coordination number 10 is retained. Thus, this macrocyclic platform appears to be a very attractive candidate to develop stable complexes with Gd(III) and other metal ions.

5. Validation of the model using a test set

To validate the structural descriptors described in the previous sections, we determined the stability constants of a series of

complexes whose stability was either not reported before (**T1–T4**) or were identified to be incorrectly reported during the course of our analysis (**T5–T6**, Fig. 12). Indeed, the stability constants reported in the literature for **T5** ($\log K_{GdL} = 10.6$) and **T6** ($\log K_{GdL} = 10.2$) are 6–7 orders of magnitude lower than those predicted by our model ($\log K_{GdL} = 15.91$) and **T6** ($\log K_{GdL} = 17.45$) [137,138]. The protonation constants of these ligands and the corresponding stability constants and pGd values are listed in Table 4. Furthermore, we incorporated into the test set a number of complexes that were reported very recently (Fig. 12) [139–144].

The calculated $\log K_{GdL}$ show a mean deviation of 1.0 $\log K$ units, with the largest deviation being observed for **T10** (2.4 $\log K$ units). We note that the values predicted for **T1–T4** are all somewhat higher than the experimental ones, which suggests that the 0.15 M ionic strength used for stability constant determination may cause a slight depression of the $\log K_{GdL}$ values. The mean deviation of the calculated pGd values is 1.00 pGd units.

The stability constants predicted for the complexes of **T5** and **T6** present an excellent agreement with the values determined using spectrophotometric titrations, confirming that the literature values obtained with pH-potentiometry were wrong. The low basicity of these ligands evidenced by their protonation constants (Table 4) results in a negligible complex dissociation at pH 2. pH potentiometry experiments can only access equilibrium processes occurring in the range $2 < pH < 13$, and thus, reliable stability constants for these systems cannot be obtained with this method. This highlights the importance of using alternative experimental methods to confirm the model used to fit potentiometric data. The correlation proposed here allows estimating the stability constants and thus the pH range in which complex dissociation is expected, therefore aiding experiment design. The stability constant of the Gd(III) complex with a heptadentate ligand containing an ethylenediamine spacer functionalized with three carboxylic acid functions and a picolinic acid one was reported recently to be $\log K_{GdL} = 17.0$ (**A79**, Fig. 13) [145]. The use of Eq [1] gives a stability constant of $\log K_{GdL} = 2 \times \Delta \log K(N) + 3 \times \Delta \log K(C) + \Delta \log K(Pic) = 19.1$. Furthermore, the experimental value is lower than that determined for the complex with H_4EDTA (**A43**), and the presence of a picolinic acid is expected to provide increased stability. The octadentate derivative containing two acetic acid and two picolinic acid arms (**A50**, $H_4OCTAPA$) provides a stability constant that is ~ 3 $\log K$ units higher than that of H_4EDTA (Fig. 13). Thus, the stability constant obtained with pH potentiometric titrations should be

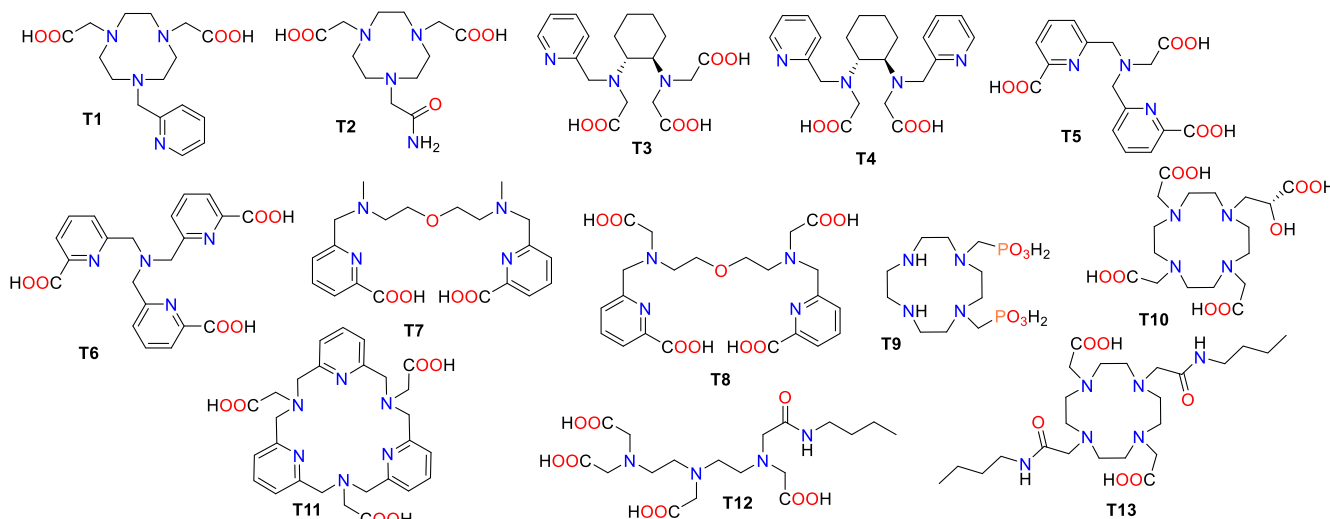
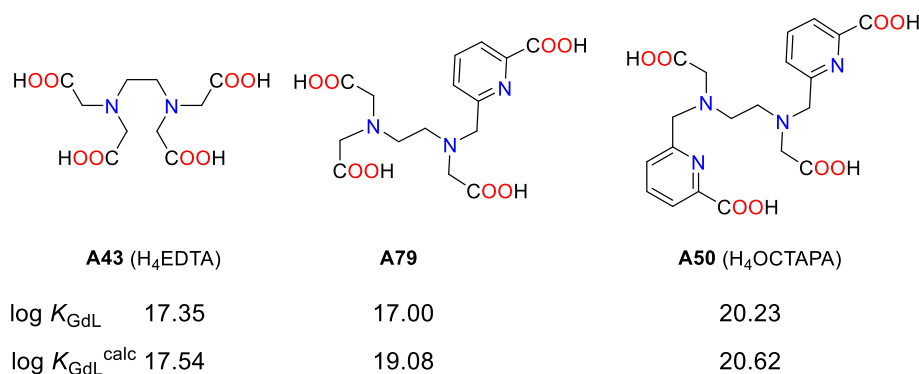


Fig. 12. Structures of the ligands used as a test set.

Table 4Ligand protonation constants (pK_a values), stability constants and pGd values determined for the Gd(III) complexes of the test set **T1–T13** and calculated $\log K_{GdL}$ and pGd values.

	pK_a	I	$\log K_{GdL}$	$\log K_{GdL}^{calc}$	pGd	pGd^{calc}	Ref
T1	11.8(2); 5.90(1); 3.91(2); 2.3(2)	0.15 M NaCl	12.55(1)	11.30	9.09	8.06	This work
T2	11.6(2); 4.4(1); 2.3(1); 2.2(2)	0.15 M NaCl	11.9(1)	10.54	7.32	7.57	This work
T3	10.12(2); 6.52(2); 3.55(2); 2.00(5)	0.15 M NaCl	14.47(5)	16.22	12.65	13.97	This work
T4	10.19(1); 6.19(1); 3.03(1); 1.14(4)	0.15 M NaCl	12.07(5)	13.84	10.21	12.50	This work
T5	7.24(3); 3.9(1); 3.1(1); 1.7(2)	0.15 M NaCl	16.2(1)	15.91	16.93	15.99	This work
T6	6.54(1); 4.78(6); 3.57(6); 2.28(6)	0.15 M NaCl	17.77(3)	17.45	18.67	18.16	This work
T7	8.72; 7.81; 3.09; 2.35	0.1 M KCl	12.02	12.66	11.09	11.91	[139]
T8	8.47; 7.63; 3.66; 2.90; 1.77	0.1 M KCl	20.50	20.78	19.94	19.91	[139]
T9	12.84; 11.20; 7.75; 4.94; 1.40	0.1 M Me ₄ NCl	19.15	18.76	10.52	11.63	[140]
T10	8.95; 8.95; 4.22; 3.74; 2.47; 1.75	0.15 M NaCl	19.26	21.67	17.10	18.01	[141]
T11	7.89; 7.35; 6.99; 2.30	0.1 M KCl	18.30	16.92	18.36	16.23	[142]
T12	9.336; 6.65; 4.09; 2.41; 2.00	2.0 M NaClO ₄	18.78	19.53	17.73	17.42	[143]
T13	10.31; 9.04; 2.86; 2.21; 1.93	1 M KCl	19.30	19.56	15.72	16.35	[144]

**Fig. 13.** Experimental and calculated stability constants for structurally related ligands that suggest that the value reported for **A79** is incorrect.

validated with the use of a complementary technique before considered as reliable.

6. Conclusions

The use of Gd(III) complexes as MRI contrast agents triggered a great amount of work to determine the thermodynamic stabilities of complexes with this metal ion. The large body of data available in the literature provides a unique opportunity to establish structure-stability correlations. It is generally accepted by the MRI community that kinetic inertness is more important than thermodynamic stability. However, thermodynamic stability may be relevant if the lifetime of the agent in the body is comparable to its dissociation half-life. Furthermore, other applications in which stable lanthanide complexation is required may benefit from the analysis presented here, which provides a straightforward tool to predict the stability constants of Gd(III) complexes and their pGd values that was obtained from the analysis of literature data. The empirical relations presented here are remarkably accurate despite known differences in experimental stability constants arising from the use of different ionic media and ionic strengths. Our approach serves to identify published stability constant data that may be incorrect. It also allows the prediction of the stability of *de novo* Gd(III) complexes. Here we see value in using the calculated stability constant to rule out synthesis of likely inferior complexes. Another feature of this approach is the quantification of different structural parameters like donor atom or chelate ring type. While certain changes like increasing chelate ring size from 5 to 6 is well known to be detrimental to complex stability for large ions like Gd (III), the current quantitative analysis shows how large this effect is and how it may compare to other structural changes. We hope that these quantitative descriptors will be useful to the novice in understanding this field.

The comparison of descriptors for $\log K_{GdL}$ and pGd is also informative and helpful in separating out contributions of ligand basicity to $\log K_{GdL}$. For instance many of the CYCLEN-based ligands have higher $\log K_{GdL}$ values than H₅DTPA-based ligands with comparable donor sets, however the pGd analysis reveals that these higher $\log K_{GdL}$ values arise from the much higher basicity of CYCLEN and that at pH 7.4 the stability of the H₅DTPA derivative may be higher. Similarly contributions to complex stability from low basicity donors like ether O or pyridyl N become apparent when comparing pGd values and comparing differences between the descriptors for $\log K_{GdL}$ vs pGd.

Here we focused on Gd(III) complexes and stability at pH 7.4. However this analysis can be readily extended in other directions. For example the same approach could be applied to other metal complexes in aqueous solution where thermodynamic stability is a key driver, e.g. radiometal applications for imaging or therapy with isotopes of Tb(III), Lu(III), Y(III), Ga(III), In(III), and Cu(II) [146,147], or for alternative MRI contrast agents based on Mn(II) or Fe(III) [148–151]. Another application is in the design of ligands to selectively chelate one metal compared to others, where comparison of the structural descriptors for one metal ion compared to another would be expected to provide guidance in ligand design that maximizes differences in stability. For such an application a lower pH to assess the pM value may also be informative.

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Declaration of Competing Interest

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ccr.2022.214606>.

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